

Assignment 12

Note: "You don't need to implement the Queue or Stack class — assume they exist and just write methods."

1. Check for Balanced Parentheses

Write a method that takes a String Input to check if a given expression containing brackets is balanced:

- Round Brackets "()"
- Curly Brackets "{ }"
- Square Brackets "[]"

Test Data:

- Input: "{ [()] }" → Output: Balanced
- Input: "{ [] }" → Output: Not Balanced
- Input: "{ (()) }" → Output: Not Balanced

2. Write a method to find the maximum element in a queue.

Test Data:

Queue: 12 35 7 98 45

Output: Maximum element: 98

3. Write a method to merge two queues alternately.

Note: If one queue runs out of elements append the remaining elements of the other queue.

Test Data:

Input:

Queue1: 1 3 5

Queue2: 2 4 6 8 10

Output:

Merged Queue: 1 2 3 4 5 6 8 10

4. Write a method to reverse the first K elements of a Queue.

Test Data:

Input: queue = 1 2 3 4 5 , k (Number of Elements to be reversed) = 3

Output: 3 2 1 4 5

5. Baseball Game using Stack

You are keeping the scores for a baseball game with strange rules. At the beginning of the game, you start with an empty record.

You are given a list of strings operations, where operations[i] is the ith operation you must apply to the record and is one of the following:

- An integer X. → Record a new score of x.
- '+'. → Record a new score that is the sum of the previous two scores.
- 'D' → Record a new score that is the double of the previous score.
- 'C' → Invalidate the previous score, removing it from the record.

Return the sum of all the scores on the record after applying all the operations.

Test Data 1:

Input: ops = ["5","2","C","D","+"]

Output: 30

Explanation:

"5" - Add 5 to the record, record is now [5].

"2" - Add 2 to the record, record is now [5, 2].

"C" - Invalidate and remove the previous score, record is now [5].

"D" - Add $2 * 5 = 10$ to the record, record is now [5, 10].

"+" - Add $5 + 10 = 15$ to the record, record is now [5, 10, 15].

The total sum is $5 + 10 + 15 = 30$.

Test Data 2:

Input: ops = ["1","C"]

Output: 0

Explanation:

"1" - Add 1 to the record, record is now [1].

"C" - Invalidate and remove the previous score, record is now [].

Since the record is empty, the total sum is 0.

Bonus Questions

1- Stack

[Problem 1 Link](#)

2- Queue

[Problem 2 Link](#)